

Mohamed Yassin

DevOps Engineer

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About Me

Fourth-year CS student with strengths in Linux administration, networking, and backend development in Node.js and TypeScript. Strong C++ and algorithms background. Experience with microservices, clustering, and Redis. Basic Python. Currently learning Kubernetes, Jenkins, Docker, and DevOps workflows.

Work Experience

- Intern** – *Egyptian Space Agency (EgSA), Earth Observation Data Department* Aug 2025 – Nov 2025
- Built backend core and frontend interfaces for the Egyptian Heritage Satellite Atlas.
 - Developed interactive maps and UI with **TypeScript + React**.
 - Integrated satellite data workflows with **Google Earth Engine**.
 - Applied remote sensing, GIS, and AI analysis for agriculture, urban growth, and water resources.

Projects

- PeerLink – Distributed File Sharing Platform** [repo](#) [🔗](#)
- Built a distributed file sharing platform with real-time transfer, load balancing, fault tolerance, and zero-downtime scaling.
 - Implemented a microservices backend with Redis cluster coordination, pub/sub routing, PostgreSQL persistence, and leader election.
 - Added native C++ acceleration for SIMD checksums and IO-heavy workloads.
 - **Tools:** TypeScript, Node.js, C++ (N-API), Redis, PostgreSQL, Prisma, Socket.IO, React, Docker.
- HoneyKube – AI-Powered Kubernetes Honeypot** [repo](#) [🔗](#)
- Developed a scalable honeypot that mimics vulnerable services using LLMs and logs attacker behavior.
 - Detects exploit scanners and stores payloads with Redis session tracking and artifact storage.
 - **Tools:** Python, Docker, Kubernetes, Redis, OpenRouter.
- GUISHell – High-Performance Remote Desktop Engine** [repo](#) [🔗](#)
- Engineered a low-latency remote desktop platform using **Electron** and **WebRTC** with a custom MJPEG-over-DataChannel streaming protocol.
 - Built a scalable **Node.js** signaling server with **SQLite (WAL mode)** persistence and **Cloudflare TURN** for NAT and firewall traversal.
 - Designed secure host consent flows, CI/CD packaging, and background auto-updates through **GitHub Releases**.
 - **Tools:** Electron, WebRTC, Node.js, SQLite, Cloudflare RTC, Electron Builder, nut.js.
- CCNA 200-301 Final Project** [repo](#) [🔗](#)
- Designed and deployed an enterprise network with VLANs, DNS, EtherChannel, STP, subnetting, and OSPF over ISP-connected infrastructure using **Cisco Packet Tracer**.

Key Achievements

- **TurboNodeIO** [🔗](#) – Built a native C++ acceleration layer for Node.js, bypassing the event loop.
- Achieved **18.95× faster average performance** than standard TypeScript.
- Reached **317× faster** 100MB reads, **56× faster** 1GB reads, **71× faster** SIMD checksums for 100KB buffers, and **45× faster** for 1GB workloads.

Technical Skills

Programming: C++ (Native N-API), TypeScript/Node.js, Bash, Python
Backend & DB: Distributed systems, microservices, WebSockets, PostgreSQL, Redis, Prisma ORM
DevOps & Cloud: Docker, Kubernetes (HPA, probes, Blue-Green), Minikube, CI/CD
Linux & Tools: System administration, automation, Nginx, Git, SSH
Networking: OSPF, VLANs, STP, EtherChannel, subnetting

Education

Modern Academy – *B.S. in Computer Science* Sept 2022 – Present

Additional Skills

Languages: Arabic (Native), English (Fluent)
Certifications: Cisco Certified Network Associate (CCNA) 200-301 Level 1
Technical: Penetration Testing, Networking, Linux
Soft Skills: Leadership, Self-Learning, Planning, Teamwork, Adaptability, Problem-Solving